

Ethan Arreguin

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EDUCATION

Virginia Tech

B.S. Aerospace Engineering — Incoming Freshman

Blacksburg, VA

Fall 2026 — Expected May 2030

Princeton High School

High School Diploma

Cincinnati, OH

August 2022 — May 2026

- Engineering coursework: Digital Electronics, Principles of Engineering, Essentials of Engineering, CAD
- Additional coursework: Computer Programming, HTML, AP Calculus, IB Physics
- Earned Fusion 360 Certified User certification through the engineering program
- GPA: 4.17 (weighted) / 3.65 (unweighted)

EXPERIENCE

StandardAero

Non-Destructive Testing (NDT) Intern

Cincinnati, OH

Summer 2025

- Rotated through five NDT methods — X-Ray, Ultrasonic, Eddy Current, Magnetic Particle, and Fluorescent Penetrant Inspection — to detect defects in critical aerospace engine components
- Applied OEM Standard Practices Manual (SPM) specifications to inspect components against precise industry tolerances
- Gained hands-on exposure to large-scale MRO shop operations, including part routing, workflow planning, and cell-based production organization
- Collaborated with engineers and technicians across NDT cells to troubleshoot inspection findings and resolve component issues

Triple V Aircraft Engine Services

Aircraft Powerplant Apprentice

Cincinnati, OH

May 2025 — Present

- Tore down aircraft engines and identified components requiring repair or replacement based on inspection standards
- Assisted in engine assembly and refurbishment to meet service standards
- Connected and operated engines on test stands
- Managed shipping, receiving, and shop organization

ENC Buildings

Real Estate Investor Intern

Cincinnati, OH

May 2022 — August 2024

- Assisted with property research and investment evaluation
- Learned the process of purchasing homes under a business entity
- Participated in residential renovation and remodeling projects
- Gained knowledge of construction methods, building systems, and basic business and tax concepts

Dewey's Pizza

Back of House Cook

Cincinnati, OH

May 2024 — August 2024

- Maintained an organized and efficient workstation in a fast-paced kitchen environment
- Assisted team members across multiple stations to maintain smooth kitchen operations

LEADERSHIP & ACTIVITIES

Robotics Club — Princeton High School

Founder & President

Cincinnati, OH

2022 — 2026

- Founded the school's robotics club from scratch; grew membership to 24 students
- Built program infrastructure and curriculum to give future students a foundation in engineering and robotics
- Led first competitive VEX robotics team (3 members); designed, programmed, and tested a disc-collecting robot
- Competed in VEX robotics individually the following year, handling all mechanical design, programming, and strategy solo
- Founded fidget toy fundraiser — 3D printed and sold toys at lunch, raising \$300+ for the chapter
- Won 1st place at University of Cincinnati FIRST LEGO Robotics Competition (2024); returning to compete in 2025
- Competed at SkillsUSA Ohio State Conference in BattleBots (2024) and VEX Robotics (2023)
- Designed a fireplace gas leak safety system that advanced to SkillsUSA State Competition (2026)

ENGINEERING PROJECTS

10+ additional projects (turbofan jet engine, combat robotics, aerodynamics research) at ethanarr.github.io

Jet Engine | Mechanical Design, Additive Manufacturing, Testing

- Independently designed, manufactured, and tested a fully functional working jet engine

Safe Descent Drone | Autonomous Systems, Embedded Programming — In Progress

- Designing a small drone device to be dropped from any height and autonomously land safely

Fireplace Gas Leak Safety System — SkillsUSA State Competition Qualifier (2026)

- Designed a safety system to detect and prevent gas leaks in the home caused by fireplace malfunctions

TECHNICAL SKILLS

Design & Manufacturing: 3D CAD Design (Fusion 360 Certified User), Additive Manufacturing / 3D Printing, Digital Electronics Design and Construction

Aerospace & Mechanical: Non-Destructive Testing (X-Ray, Ultrasonic, Eddy Current, Magnetic Particle, Fluorescent Penetrant), Aircraft Engine Maintenance and Assembly, Mechanical Troubleshooting and Repair, Shop Organization and Inventory Management

Programming: Basic Programming and HTML

Languages: English (Native), Spanish (Native)